

PCB Documentation

AGH Space Systems - Szymon Hamownia

Mikołaj Sala, Piotr Słonka

July 8, 2026

Contents

1	Project Overview	3
2	Specifications & Requirements	4
3	Functional Description	4
3.1	Power Over Ethernet (PoE) Stage	5
3.2	Ethernet Data Stage	5
3.3	Main Computing Unit	5
4	Schematic Breakdown	6
4.1	PoE Input Section	6
4.2	DC/DC Converter	6
4.3	Ethernet PHY Section	7
5	PCB Layout	7
5.1	Stackup and layers	8
5.2	PCB Layout Notes	8
5.3	PCB Layout View	9
6	Bill of Materials (BOM)	10
7	Assembly Instructions	10
8	Testing & Validation	10
9	Revision History	10

Project Overview

This document describes the design, assembly, and testing procedures for the Szymon Hamownia Master Board (MB) — a core element of the bi-propellant Rocket Engine Test Stand (RETS) control system. The MB uses the Ethernet communication protocol, chosen for its simplicity, widespread adoption, and ease of integration. This choice allows straightforward interfacing with other Master Boards and the main server station via a managed switch.

The integration of Power over Ethernet (PoE) enables both power and data to be sent over a single cable, reducing wiring complexity and improving safety in hazardous environments such as those near operating sounding rocket engines.

Functionally, the MB handles data acquisition and command transmission to various subsystems via the communication protocols used by connected sensors and servomechanisms. It also bridges these devices into the broader network necessary for real-time control and monitoring of the RETS.

The board is designed to operate reliably under challenging conditions, including vibrations, moisture, and temperature variations encountered during rocket engine testing.

The goal of this document is to provide clear guidelines for future engineers to reproduce, repair, and improve the design, ensuring the continued reliability and adaptability of the RETS electronics.

Key Features:

- external Ethernet (100Base-TX) communication
- Power over Ethernet (PoE Type 1)
- ARM Cortex - M4 32-bit RISC MCU (STM32F439ZI6U)
- internal SPI, I²C, USART, CAN communication protocols
- 48 V input, 5 V output @ up to 10 W
- Integrated magnetics RJ45 connector

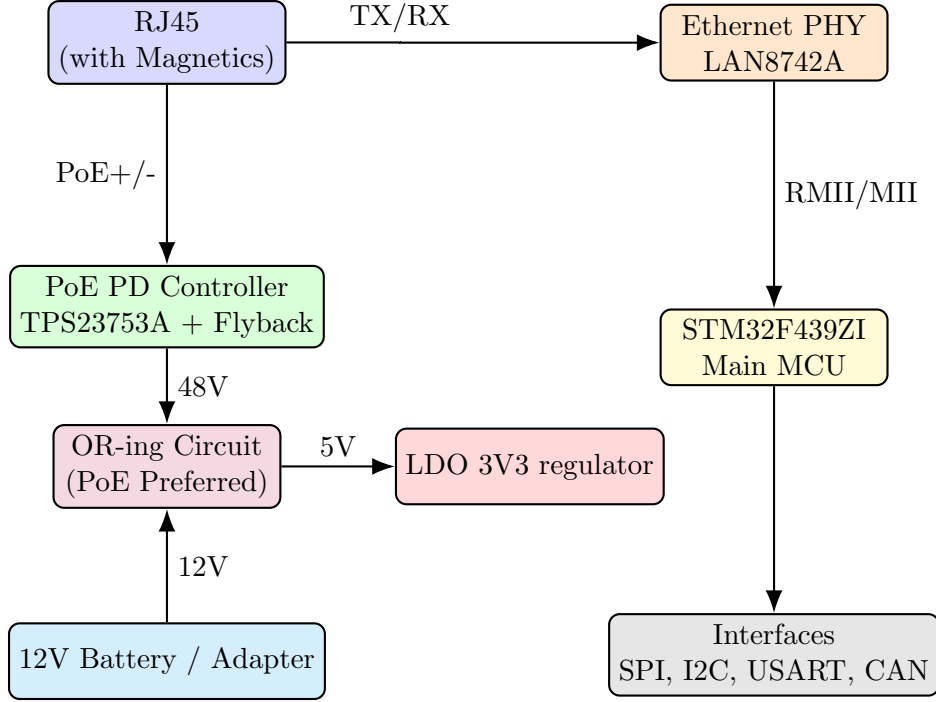


Figure 1: High-level system block diagram with PoE and 12V adapter input (PoE preferred).

Specifications & Requirements

Parameter	Value
PoE Input Voltage	37–57 V (PoE Type 1 standard)
Auxiliary Input Voltage	12 V
Output Voltage	5 V @ 2 A
Output Power	Up to 10 W
Communication	100 Mbps Ethernet
Operating Temperature	−20 °C to 70 °C
PCB Size	100 mm × 100 mm
PCB Layers	4 layers

Table 1: Electrical and mechanical specifications

Functional Description

The board consists of three main functional subsystems:

1. **PoE Power Delivery Unit:** TPS23753APWR controller with NA6223-ALD transformer for 48 V to 5 V conversion.
2. **Ethernet Data Stage:** handled by LAN8742A PHY.
3. **Main Computing Unit:** using STM32F439ZI6U microcontroller for data handling

Power Over Ethernet (PoE) Stage

This project uses the Power over Ethernet (PoE) protocol — specifically Type 1, Mode B, compliant with the IEEE 802.3af standard. In this configuration, power is delivered over the spare pairs in the Ethernet cable, while data is transmitted over the remaining pairs.

Type 1 PoE was selected for its simplicity, wide compatibility, and ability to deliver approx. 12.95 W available at the device after cable losses, which is sufficient for this application.

This approach reduces cabling complexity and increases safety in harsh environments such as RETS, where minimizing extra connections helps reduce the risk of electrical hazards and mechanical failures.

Pin (RJ45)	Signal
1	TX+ (Transmit Data +)
2	TX- (Transmit Data -)
3	RX+ (Receive Data +)
4	V+ (PoE DC +)
5	V+ (PoE DC +)
6	RX- (Receive Data -)
7	V- (PoE DC -)
8	V- (PoE DC -)

Table 2: Pinout of Ethernet connector

Ethernet Data Stage

Describe PHY connections, differential pair routing

All differential pairs were routed with precise length matching, maintaining a maximum mismatch of 0.1 mm to ensure timing alignment and signal integrity. The routing also adheres to Ethernet clearance requirements, with a minimum spacing of 0.508 mm, which corresponds to the most stringent constraint typically applied to 1 Gbps connections, with total routed lengths $TX = 99.564$ mm and $RX = 98.875$ mm

Main Computing Unit

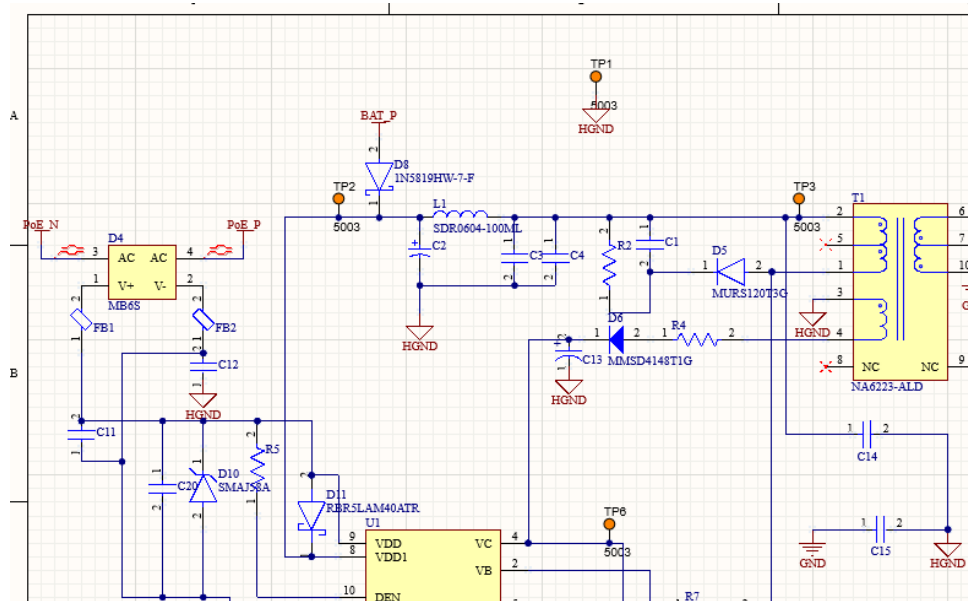
The table below consists of pins of the MCU mapped with their usage in the project.

Pin (MCU)	Usage	Pin (MCU)	Usage	Pin (MCU)	Usage
PA8	USART1 – CK	PE2	SPI4 – SCK	PD1	CAN1 – TX
PA9	USART1 – TX	PE4	SPI4 – NSS	PB5	I ² C1 – SMBA
PA10	USART1 – RX	PE5	SPI4 – MISO	PB6	I ² C1 – SCL
PA11	USART1 – CTS	PE6	SPI4 – MOSI	PB7	I ² C1 – SDA
PA12	USART1 – RTS	PD0	CAN1 – RX	—	—

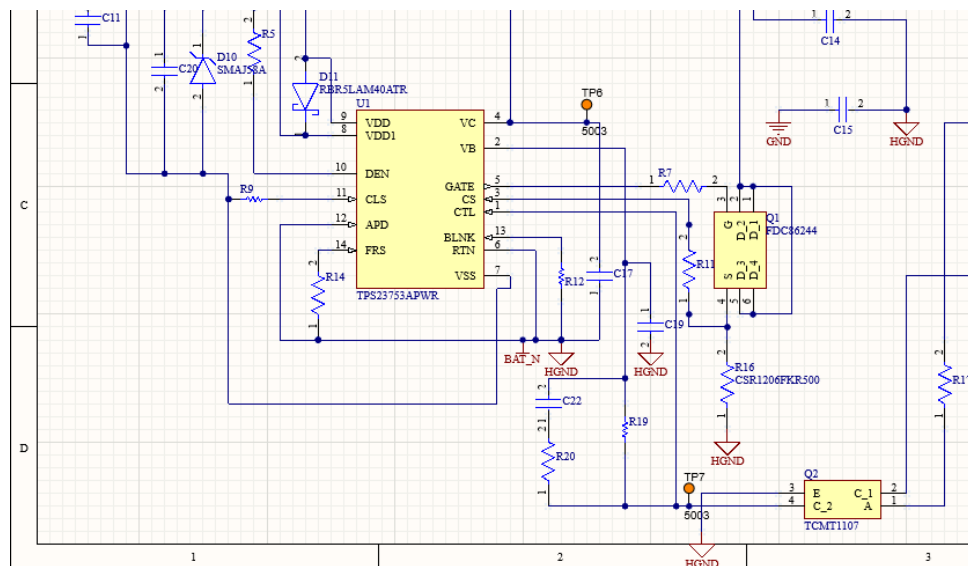
Table 3: Table of used pins in MCU

Schematic Breakdown

PoE Input Section



DC/DC Converter

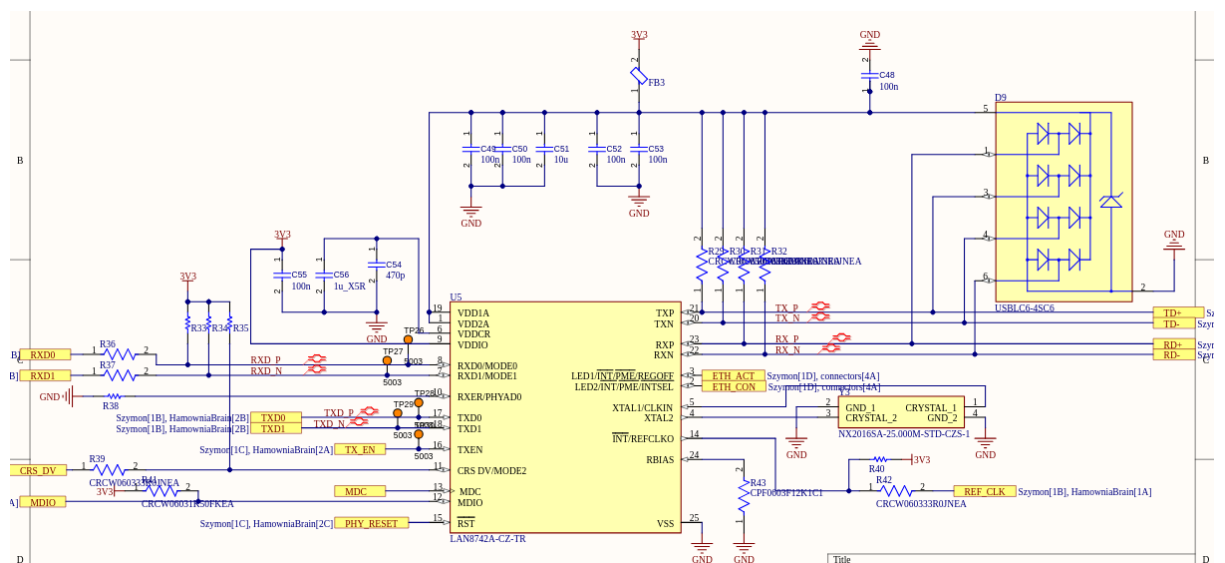


The DC–DC stage is implemented as an **isolated flyback converter**, controlled by the TPS23753A PWM controller. The flyback topology is particularly well suited for Power over Ethernet (PoE) applications for several reasons.

First, it provides **galvanic isolation** between the Ethernet line side and the powered circuitry, which is a strict requirement of the IEEE 802.3 PoE standard. Second, the topology tolerates the wide input voltage range of **37–57 V** typical for PoE, without the need for additional conversion stages. Flyback converters are also efficient in the **low-to-mid power range** (up to ~15 W), making them a good match for PoE Type 1 devices.

In addition to the basic flyback structure, the design incorporates several measures to ensure safe and reliable operation. **Clamp and snubber networks** are used across the transformer primary and MOSFET drain to suppress voltage spikes generated by leakage inductance during switching. This reduces stress on the switching transistor and minimizes high-frequency ringing that could otherwise propagate as electromagnetic interference (EMI).

Ethernet PHY Section



7

Stackup and layers

	Top Overlay		Overlay				
	Top Solder	Solder Resist	Solder Mask	0.4mil	3.5		
1	Top Layer	CF-004	Signal	1.378mil			1oz
	Dielectric 2	PP-7628	Prepreg	8.284mil	4.1	0.02	
2	Power	CF-004	Signal	1.378mil			1oz
	Dielectric 1	FR-4	Core	41.929mil	4.8		
3	Ground	CF-004	Signal	1.378mil			1oz
	Dielectric 3	PP-7628	Prepreg	8.284mil	4.1	0.02	
4	Bottom Layer	CF-004	Signal	1.378mil			1oz
	Bottom Solder	Solder Resist	Solder Mask	0.4mil	3.5		
	Bottom Overlay		Overlay				

Figure 2: PCB Stackup

- For Ethernet working as intended it is crucial to use prepreg-7628, because the design provides satisfactory impedance on differential pairs.
- L1 is reserved especially for relatively slow signals however as seen in the project sometimes the differential pair of Ethernet signal occur on this plane (it is acceptable to some extend), however exceeding this can put the risk of cross-talks or EMI.
- L2 consist only plains of supply voltage of 5 V at the region of LDO, and 3.3 V for most of the logic circuits such as MCU and PHY.
- L3 main goal is to provide grounding for every part of PCB, as well as shielding from power plane for fast signals such as Ethernet. This plane is divided into 3 main sections: ETH.GND for RJ45 magnetics shielding, HGND for high power PoE circuit and GND for logic/signal circuits. Remember to use stitching vias (hole size: 30mil, diameter 60mil, grid 80mils)
- L4 layer should consist only fast signal routs, however it is acceptable as before to switch layers of signals temporarily.

PCB Layout Notes

- Differential pairs 100 Ω impedance as well as odd impedance of 50 Ω with $\pm 10\%$, their length should not be greater than 6000mil or 15.24 cm.
- Ground plane separation between PoE and the rest of signals is crucial to maintain integrity neither component should cross ground planes, beside those three: transformer (T1), opto-coupler (Q2) and one capacitor (C15).
- Keep PoE high-voltage traces away from low-voltage logic and differential pairs if necessary use shielding vias.

PCB Layout View

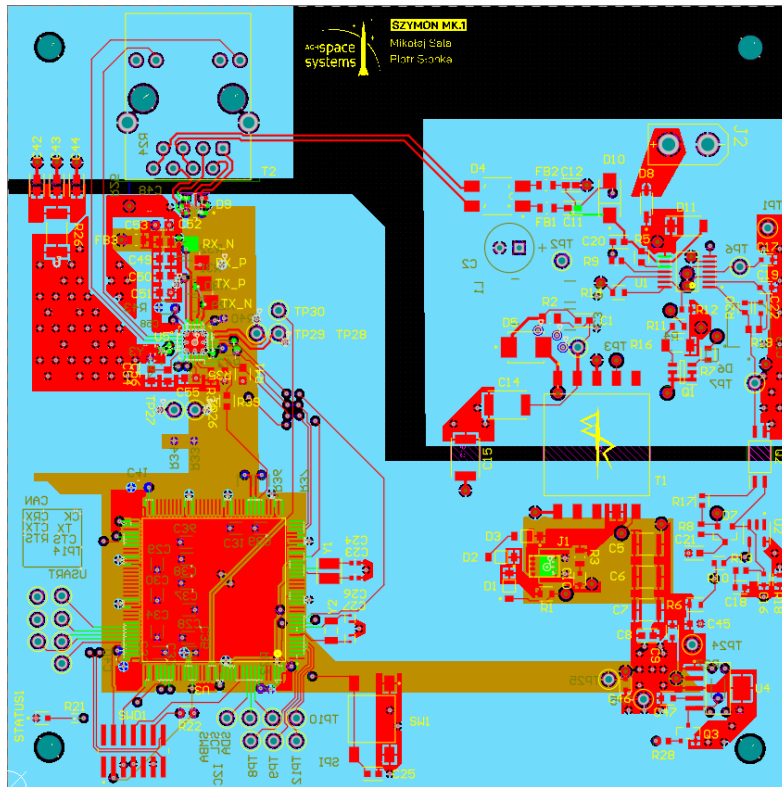


Figure 3: Top view of the PCB layout

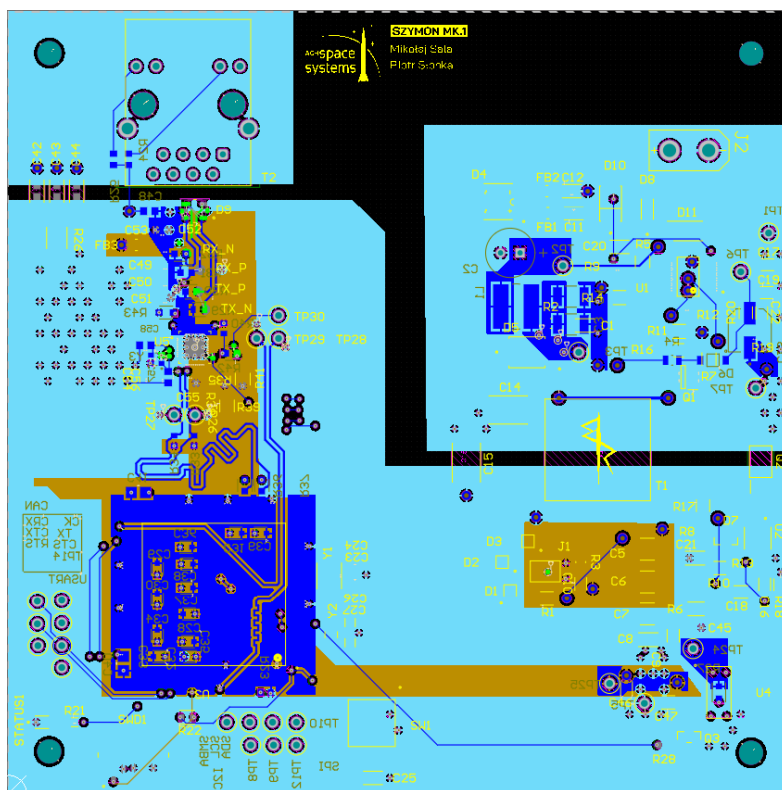


Figure 4: Bottom view of the PCB layout

Bill of Materials (BOM)

Up-to-date Bill of Materials is available as an appendix 1 to this documentation.

Assembly Instructions

1. Solder all SMD components first (reflow recommended).
2. Hand-solder through-hole components.
3. Verify polarity of electrolytic capacitors and diodes.

Testing & Validation

Revision History

Date	Version	Changes
2025-08-09	1.0	Initial release
2025-08-11	1.1.0	Auxiliary Battery supply added
2025-08-20	1.1.1	Fixed ETH to be compatible with Nucleo F439ZIU
2025-09-30	1.2	Fixes expected: footprint mismatch, TP designators, unitary BtB connectors with Slaves, remake DC/DC to 12V output

References

- TPS23753A Datasheet – Texas Instruments
- LAN8742A Datasheet – Microchip
- STM32F439ZI Datasheet - STMicroelectronics
- REG103GA-3.3 Datasheet - Texas Instruments
- Ethernet PHY Design Checklist - Texas Instruments
- IEEE 802.3af PoE Standard